

YASH RANJAN

3800 SW 34th St, Gainesville, FL 32608

352-709-3789 [◇ yashranjan@ufl.edu](mailto:yashranjan@ufl.edu) [◇ linkedin.com/ranjanyash54/](https://www.linkedin.com/ranjanyash54/)

WORK EXPERIENCE

Research Assistant, University of Florida

05/2024 - present

- Building data driven simulation in the latent space using Diffusion models for scientific simulation applications such as weather prediction and CFD simulations.
- Working on world models to bridge the gap between deep-learning models used in self-driving domain and the explainability required for smarter public infrastructure.
- Built a traffic simulation model using Transformers architecture to model the multi-agent (vehicles, pedestrians, signal etc) interactions and a data harmonization framework to combine data from multiple scenes and generalize on held out scenes showing improvements of 36% compared to model trained on a single scene with strong zero shot results.
- Developed a Variational Auto-Encoders based trajectory prediction model that combined curvilinear coordinates with intersection related controls and performed 17% better on displacement metrics than strong baseline.
- Worked on a novel “simulation-in-the-loop” framework to evaluate trajectory prediction models on multi agent simulation tasks and proposed a set of traffic engineering based metrics: unsafe deceleration, lane following and red-light violations.
- Maintained and operated the end-to-end computer vision data pipeline for traffic-intersection video analytics, configuring YOLOv7 object detection and DeepSORT tracking to deliver high-quality trajectory data for near-miss conflict analysis, maintaining uptime of over 99%.

Software Intern, Nutanix, San Jose, CA

05/2023 - 8/2023

- Engineered an efficient messaging bus for security policy framework to send and receive security policies between hypervisor and management VM and stress tested it to a 1 million+ policies beating the existing pipeline by over 10 percent.
- Created a Front-End page using React-JS framework with dynamic page refresh to show the security policies applied on containers running on the hypervisor with controls for adding sub-net masks and IP addresses.

Member of Technical Staff, VMware, Bangalore, India

07/2020 - 07/2022

- Worked on the Edge-Routing team developing routing protocols for the open-stack routing framework Free Range Routing (FRR).
- Designed and implemented data structures for transaction-driven asynchronous CRUD operations for management of network routing information without compromising the transaction consistency.
- Contributed features such-as commit-rollback (reversing the transaction), asynchronous read operations (for fetching large operational data without delaying other transactions), and wildcard query resolution (similar to *find* command in GNU)

Undergraduate Researcher, BITS-Pilani, India

08/2018 - 05/2019

- Created and scaled community detection, feature extraction and event prediction pipeline for more than 64K nodes and 1.5M edge social network dataset using Apache Spark framework with modularity of 0.91.
- Implemented an ensemble learning algorithm that combined the output of standard ML algorithms like Gradient Boosting, Random Forest, SVM, for event classification with a test accuracy of 82%.

EDUCATION

University of Florida

08/2022 - present

- PhD - Computer Science

- M.Sc.(Hons.) - Computer Science

Thesis: Probabilistic Modelling of driving behaviour at a traffic intersection

Advisor: Dr. Sanjay Ranka, Dr. Anand Rangarajan

BITS Pilani, India
- B.E.(Hons.) Computer Science

08/2016 - 06/2020

TECHNICAL EXPERTISE

Languages	Python, C, C++, GoLang, Java, JavaScript, Erlang
Machine Learning	PyTorch, CUDA, cuDNN, Apache Spark, Hadoop, YOLOv7, DeepSORT
Tools	SUMO, CARLA, TensorBoard, Weights & Biases, Git, GitHub, Docker

PREPRINTS AND PUBLICATIONS

- [1] **Ranjan Y**, Sengupta R, Rangarajan A, Ranka S. *PhaseShift: Topology-Aware Data Harmonization and Model Consolidation Across Signalized Intersections* (Submitted, In-Review) <https://arxiv.org/abs/2608.25275>
- [2] **Ranjan Y**, Sengupta R, Rangarajan A, Ranka S. *A Generative Model for Closed-Loop Microsimulation of Signalized Intersections* (Submitted, In-Review) <https://arxiv.org/abs/2606.23588>
- [3] **Ranjan Y**, Sengupta R, Rangarajan A, Ranka S. *Enactor: From Traffic Simulators to Surrogate World Models*. 2026 International Conference on Vehicle Technology and Intelligent Transport Systems (Accepted) <https://arxiv.org/abs/2603.18266>
- [4] **Ranjan Y**, Sengupta R, Rangarajan A, Ranka S. *IntTrajSim: Trajectory Prediction for Simulating Multi-Vehicle driving at Signalized Intersections*. 2025 IEEE International Conference on Intelligent Transportation Systems (Accepted) <https://arxiv.org/abs/2506.08957>
- [5] **Ranjan Y**, Sengupta R, Rangarajan A, Ranka S. *Evaluating Generative Vehicle Trajectory Models for Traffic Intersection Dynamics*. 2025 Pacific Asia Conference on Knowledge Discovery and Data Mining (Special Session) (Accepted). <https://arxiv.org/abs/2506.08963>
- [6] Wu A, **Ranjan Y**, Sengupta R, Rangarajan A, Ranka S. *A Data-driven Approach for Probabilistic Traffic Prediction and Simulation at Signalized Intersections*. 2024 IEEE Intelligent Vehicles Symposium (IV) (Accepted). <https://doi.org/10.1109/IV55156.2024.10588424>
- [7] Rajita B.S.A.S, **Ranjan Y**, Umesh C.T, Panda S. *Spark-Based Parallel Method for Prediction of Events*. Arab J Sci Eng 45, 3437–3453 (2020) (Accepted). <https://doi.org/10.1007/s13369-020-04437-2>

REFERENCES

Dr. Sanjay Ranka Distinguished Professor Dept. of Computer Science University of Florida Email: ranka@cise.ufl.edu	Dr. Anand Rangarajan Professor Dept. of Computer Science University of Florida Email: anand@cise.ufl.edu
Dr. Subhrakanta Panda Associate Professor Department of Computer Science BITS Pilani Hyderabad Campus Email: spanda@hyderabad.bits-pilani.ac.in	Mr. Naveen T Manager VMware India Email: naveen@vmware.com